Project Protocol:

Quantifying Cellular Damage and Growth Dynamics in Mouse Cells

Group 7 Information:

Name	Units	Role
Ella Li	9	Jack of All Trades (did random stuff) 🤡 Experimental design, led experiment, protocol, presentation
Kenneth Lu	9	Microscope GOAT 🐐
Avery Sun	12	Jack of All Trades 2 (did random stuff) 🤡Experimental design, led experiment, protocol
Zhihao Yuan	9	Computational GOAT 🐐
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Introduction:

Among the many environmental factors that affect cells, light plays a particularly important role because it directly interacts with cellular structures and its effects depend strongly on wavelength. High-energy light, such as ultraviolet (UV), can damage DNA and disrupt cellular function, while lower-energy light, such as red light, can influence processes like cell growth and repair (Bahadoran et al., 2020). Despite these known differences, the extent to which light exposure at different wavelengths quantitatively affects cell growth and survival remains an important question.

In this experiment, we investigate how different wavelengths of light, in particular, UV light and red light, affect cellular damage and growth dynamics in 3T3 mouse fibroblast cells. 3T3 cells are a commonly used fibroblast cell line derived from mouse embryos and are widely used in studies of cell growth, wound healing, and cellular response to environmental stressors. Fibroblast cells are particularly useful because they are adherent and grow as flat, uniform layers, making them well suited for imaging and quantitative analysis (ATCC, n.d.). Additionally, cell culture systems provide a controlled and easily manipulated environment, allowing for consistent comparison of how external factors, such as light exposure, affect cellular behavior.

The purpose of this project is: (1) to determine whether and how light wavelength influences cell viability and growth, and (2) to quantitatively characterize these effects using both experimental and computational approaches. Understanding these relationships has broader significance in both public health and biomedical applications. For example, excessive UV exposure is a known risk factor for tissue damage and disease, while controlled light exposure (such as red-light therapy) is increasingly being explored for therapeutic purposes, including wound healing and tissue regeneration (Bahadoran et al., 2020). By investigating these effects in a controlled

experimental setting, this study aims to contribute to a clearer understanding of how light interacts with living cells.

Safety Concerns:

1. **UV Light Exposure:** Ultraviolet light poses a significant hazard, as it can damage skin and especially eyes with even brief exposure. Although the UV source is enclosed within the biosafety cabinet, it is important to ensure that the UV light is only used when the hood sash is properly closed so that no direct exposure occurs. Avoid placing hands or skin inside the hood while UV light is active, and wait for the light to fully turn off before retrieving cells between experimental rounds (Centers for Disease Control and Prevention [CDC], n.d.).
2. **Cell Contamination:** Sterile technique must be maintained when handling 3T3 cells to prevent contamination. All work involving the cells, such as pipetting, should be performed inside the biosafety hood with the laminar airflow turned on to maintain a sterile environment. Use a new, sterile pipette tip for each task. Ensure the lid of the plate is kept on when microscoping. Contamination could compromise experimental results and affect cell viability.
3. **General Lab Safety:** Standard lab safety practices should be followed when using equipment such as microscopes, pipettes, and incubators. Appropriate protective equipment, including lab coats, gloves, and face shields, should be worn at all times to minimize exposure to biological materials and UV light. Care should be taken to avoid spills, handle equipment properly, and dispose of all biological materials in designated biohazard waste bins.

Experimental Overview:

To investigate how light wavelength affects cell growth and damage, we used a scratch assay with cultured 3T3 mouse fibroblast cells. The cells were first grown in multi-well plates until they formed a mostly confluent layer. A scratch was then introduced into each well using a sterile pipette tip, creating a cell-free gap that mimics a wound . This allowed us to observe how cells migrated and proliferated to fill the gap over time (University of Virginia, Department of Otolaryngology, 2020).

The experiment included three treatment conditions: UV light exposure, red light exposure, and a no-light control. For the control condition, wells were covered with aluminum foil to prevent light exposure, providing a baseline for normal cell growth and scratch closure. For the UV and red light treatment groups, an incremental exposure design was used to generate multiple exposure durations within each light condition. All wells assigned to a given light treatment were initially exposed at the same time. After 10 seconds, the well assigned to the 10-second condition was covered with aluminum foil to stop further exposure. The remaining wells continued to be exposed, and the 1-minute well was then covered after an additional 50 seconds. This process was repeated for the 5-minute and 10-minute conditions, so that each well received a different total exposure time while keeping all other conditions as consistent as possible (National Center for Biotechnology Information [NCBI], 2023).

Images were collected immediately after scratching to establish a baseline and again after each treatment using consistent microscope settings. After exposure, the plates were returned to the incubator so the cells could continue to recover and grow. Additional images were taken after 24 hours to assess the longer-term effects of light exposure on cell recovery and growth.

Materials Needed:

- ☐ 3t3 Mouse Fibroblast Cells
- ☐ At least 5 x 12 multiwell-plate (more rows better)
- ☐ Cell Culture Media
- ☐ Biosafety Cabinet with UV light
- ☐ Red Light source (LED)
- ☐ Aluminum Foil
- ☐ 10 µL pipette tips (for scratching)
- ☐ Timer
- ☐ Incubator
- ☐ Microscope
- ☐ PPE (gloves, labcoat, face shield)
- ☐ Biohazard waste beaker

Protocol:

A. Cell Preparation

1. If not already prepared, seed 3T3 cells into multi-well plates and incubate until a confluent monolayer is achieved.
2. Remove 3T3 cells from the incubator and inspect each well under the microscope to confirm that the cells are healthy and form a mostly confluent layer.

B. Scratch Assay Setup

3. Using a sterile pipette tip, create a vertical scratch in each well by dragging the tip up and down through the cell layer twice.
    a. Apply consistent pressure and keep the pipette tip perpendicular to ensure that the scratches are as uniform as possible across wells.
4. Inspect each well and carefully re-scratch if there does not seem to be a visible scratch.
5. Take baseline images of each well using the microscope under consistent magnification and imaging settings, and label each image according to its corresponding well.

C. Assigning Experimental Conditions

6. Assign conditions:[1]
    a. Designate separate wells for the no-light control, red light treatment, and UV light treatment.
    b. Within each light treatment group, assign wells to the different total exposure durations (e.g., 10 seconds, 1 minute, 5 minutes, and 10 minutes).

[1] See Figure 1. on the next page for an example.

Tips:

- Arrange the wells so that those belonging to the same light treatment are grouped together and can be easily covered with aluminum foil as exposure times are reached.

- Leave sufficient spacing between different treatment groups when possible in order to reduce unintended light exposure between conditions.

- Label all wells clearly before beginning the experiment so that each well can be matched to its corresponding treatment condition and exposure duration during imaging and analysis.

Figure 1: Sample Plating Used for Experiment

	1	2	3	4	5	6	7	8	9	10	11	12
	NLC	NLC	R10s	R1m	R5m	R10m	NA	NLC	UV10s	UV1m	UV5m	UV10m
A							NA					
B							NA					
C							NA					
D							NA					
E							NA					

Caption: Sample plating arrangement used for the experiment. Column 1 served as the overall no-light control, while Column 2 served as the no-light control for the red light treatment and Column 8 served as the no-light control for the UV light treatment. Columns 3-6 corresponded to incrementally increasing red light exposure durations (10 seconds, 1 minute, 5 minutes, and 10 minutes), while Columns 9-12 corresponded to similarly incrementally increasing UV light exposure durations. Column 7 was unused to provide additional spacing between the red and UV treatment groups to reduce light contamination.

D. Light Exposure (Incremental Design)

7. **Remove the lid** and cover control wells completely with aluminum foil to block all light exposure. Keeping the lid on would cause the lid to absorb most of the light, rendering the light treatment useless.
    a. Keep all wells in the biosafety hood or designated exposure area before beginning treatments.

UV Light Condition:

8. Place all wells assigned to the UV condition under the UV light source at a fixed distance.
9. Turn on the UV light and begin timing exposure.

10. After **10 seconds**, turn off the UV light, replace the lid and extract the plate. Image, then cover the wells designated for the 10-second condition with aluminum foil to stop further exposure.
11. Return plate to the biosafety cabinet to the same position as before and continue exposing the remaining wells.
12. After a total of **1 minute (50 additional seconds)**, turn off the UV light, replace the lid and extract the plate. Image, then cover the wells designated for the 1-minute condition, repeat step 11.
13. After a total of **5 minutes (4 additional minutes)**, turn off the UV light, replace the lid and extract the plate. Image, then cover the wells designated for the 5-minute condition, repeat step 11
14. After a total of **10 minutes (5 additional minutes)**, turn off the UV light, replace the lid and extract the plate. Image the final wells.

Red Light Condition:

15. Repeat Steps 8-14 using the red light source for wells assigned to the red light condition.

    Note: In our case, we used a red LED enclosed in a dark box to reduce light contamination. The box was placed in the biosafety cabinet with the airflow turned on whenever the lid was removed from the plate.

E. Post-Treatment Handling

16. Once all exposures are complete and all images of immediate exposure have been taken, return the plates to the incubator. Allow the cells to recover and continue growing for 24 hours.

F. Final Imaging

17. After 24 hours, remove the plates from the incubator.
18. Take images of each well using the same microscope settings as before. Ensure the same region of the scratch is imaged for consistency.

Imaging Tips – *For both treatment and final imaging:*

- ➢ When taking images after baseline, try to center each well in the same way as in the baseline image from Step 5 to maintain consistency across time points.

- ➢ Label images immediately with the well ID, treatment condition, exposure duration, and time point to avoid confusion during analysis.

- ➢ Use the same microscope magnification and settings for all images so that wells can be compared consistently.

Computational Approach:

The protocol should be a detailed document that describes your experimental and computational approach and methods. It should include the following:
1. Group Members and units for each
2. Group Member roles
3. Introduction/Purpose
4. Safety Concerns
5. Experimental Setup/Overview (include controls)
6. Materials Needed
7. Specific and Detailed Protocol (your protocols should be more specific than the protocols I give you!).
8. Computational approach (describe algorithms in depth).

We will not enforce a page min./max., but the level of detail will be heavily considered in grading. Your protocol should be detailed enough that another student in this class could independently follow it to replicate your experiments. Grading: 5 points on March 10, 25 points on April 21, graded on detail, completeness, and understanding of experimental and computational approaches

References

ATCC. (n.d.). *CRL-1658*. https://www.atcc.org/products/crl-1658

Bahadoran, Z., Mirmiran, P., Kashfi, K., & Ghasemi, A. (2020). The principles of biomedical scientific writing: Citation. *International Journal of Endocrinology and Metabolism, 18*(2), e102622. https://pubmed.ncbi.nlm.nih.gov/3186933/

Centers for Disease Control and Prevention. (n.d.). *UV radiation*. https://www.cdc.gov/radiation-health/features/uv-radiation.html

National Center for Biotechnology Information. (2023). *Article PMC9794538*. PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC9794538/

University of Virginia, Department of Otolaryngology. (2020). *Scratch assay protocol*. https://med.virginia.edu/otolaryngology/wp-content/uploads/sites/244/2020/06/Scratch-Assay-protocol.pdf